Indian Premier League IPL - Analysis

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SUBJECTIVE QUESTIONS

1. **How does the toss decision affect the result of the match? (Which visualizations could be used to present your answer better?) And is the impact limited to only specific venues?**

A. **Approach :**

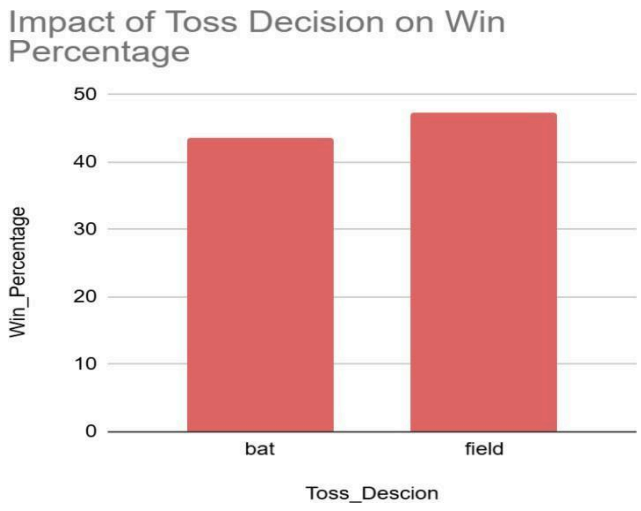
To analyze whether the toss decision (batting or fielding first) significantly influences match outcomes, and to assess if this impact varies across different venues.I have Written this query.

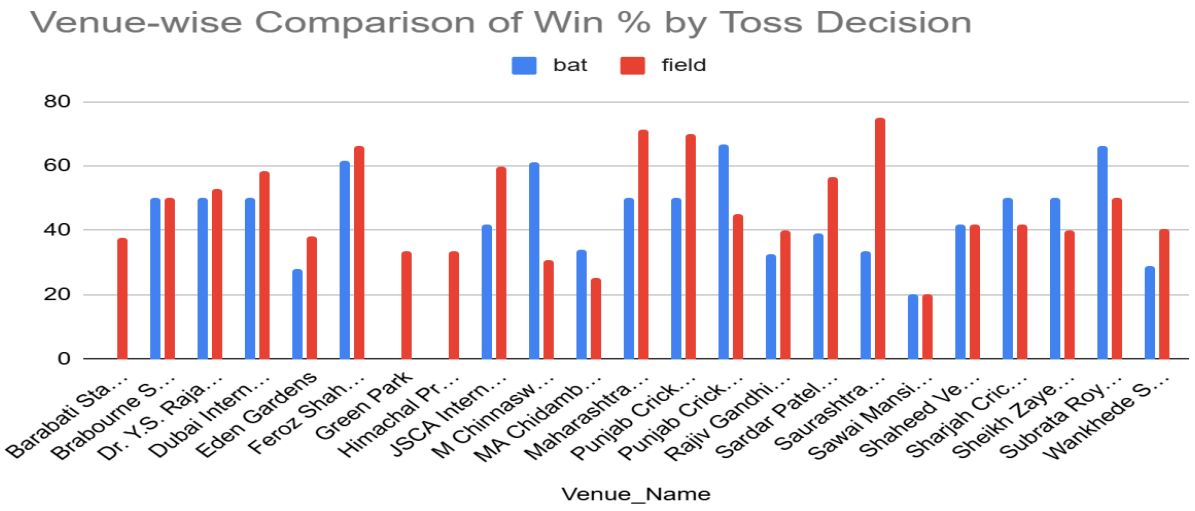
```
-- QUERY --
WITH MATCHVENUE AS (
        SELECT m.Match_Id , v.Venue_Name, m.Toss_Winner, m.Match_Winner ,
            td.Toss_Name AS Toss_Descion ,
            t1.Team_Id AS Team1_Id , t1.Team_Name AS Team1_Name ,
            t2.Team_Id AS Team2_Id , t2.Team_Name AS Team2_Name
        FROM matches m
        JOIN venue v ON m.Venue_Id = v.Venue_Id
        JOIN toss_decision td ON td.Toss_Id = m.Toss_Decide
        JOIN team t1 ON m.Team_1 = t1.Team_Id
        JOIN team t2 ON m.Team_2 = t2.Team_Id ) ,
FULLTABLE AS (
        SELECT Match_Id , Venue_Name , Toss_Winner , Match_Winner ,Toss_Descion , Team1_Id AS Team_Id , Team1_Name as Team_Name
        FROM MATCHVENUE
        UNION ALL
        SELECT Match_Id , Venue_Name , Toss_Winner , Match_Winner ,Toss_Descion , Team2_Id AS Team_Id , Team2_Name as Team_Name
        FROM MATCHVENUE )
SELECT Team_Name , Venue_Name ,
    COUNT(DISTINCT Match_Id) AS Played_Matches ,
    COUNT(DISTINCT CASE WHEN Team_Id = Toss_Winner THEN Match_Id END ) AS Toss_Wins ,
    Toss_Descion ,
    COUNT(DISTINCT CASE WHEN Team_Id = Match_Winner THEN Match_Id END ) AS Match_Wins ,
    ROUND(COUNT(DISTINCT CASE WHEN Team_Id = Match_Winner THEN Match_Id END )*100 / COUNT(DISTINCT Match_Id) , 2) AS Win_Percentage
FROM FULLTABLE
GROUP BY Team_Name , Venue_Name , Toss_Descion
ORDER BY Team_Name , Venue_Name ;
```

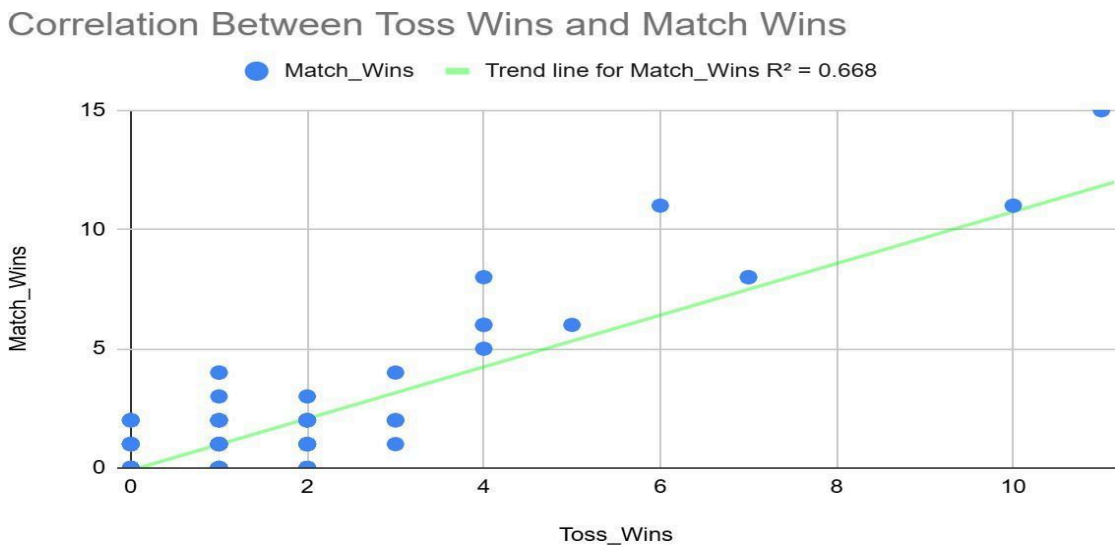
OUTPUT :

Team_Name	Venue_Name	Played_Matches	Toss_Wins	Toss_Descion	Match_Wins	Win_Percentage
Chennai Super Kings	Barabati Stadium	1	1	field	0	0.00
Chennai Super Kings	Brabourne Stadium	1	1	field	1	100.00
Chennai Super Kings	Dubai International Cricket Stadium	1	0	bat	1	100.00
Chennai Super Kings	Dubai International Cricket Stadium	1	0	field	1	100.00
Chennai Super Kings	Eden Gardens	2	0	bat	1	50.00
Chennai Super Kings	Eden Gardens	3	1	field	0	0.00
Chennai Super Kings	Feroz Shah Kotla	2	2	bat	2	100.00
Chennai Super Kings	Feroz Shah Kotla	1	1	field	1	100.00
Chennai Super Kings	JSCA International Stadium Complex	3	2	bat	2	66.67
Chennai Super Kings	JSCA International Stadium Complex	2	1	field	1	50.00
Chennai Super Kings	M Chinnaswamy Stadium	3	2	field	2	66.67
Chennai Super Kings	MA Chidambaram Stadium, Chepauk	11	7	bat	8	72.73
Chennai Super Kings	MA Chidambaram Stadium, Chepauk	4	1	field	4	100.00

VISUALIZATIONS :







INSIGHTS :

- Teams winning the toss and choosing to field first generally had a higher Win %, indicating a clear advantage while chasing.
- The impact of the toss decision isn't uniform — certain venues show strong bias (e.g., batting first works better on dry pitches, while chasing dominates on dew-prone grounds).
- Win percentage after losing the toss remains reasonably competitive, showing that match strategy matters more than just toss luck.
- The correlation between Toss Wins and Match Wins (R^2 from scatter plot) is moderate, proving toss is a contributing factor, not a decisive one.

RECOMMENDATIONS :

- Teams should analyze venue-specific trends before deciding to bat or field; adopt dynamic toss strategies instead of fixed preferences.
- At venues where dew or chasing advantage is high, teams should opt to field first when they win the toss.
- For batting-friendly venues, teams should bat first and aim to set a high total — data supports stronger results when batting first there.
- Analysts and coaches should continuously update venue-wise toss data to refine future match strategies and pre-match planning.

2. Suggest some of the players who would be the best fit for the team.

A. Approach :

To identify the most suitable players for the team by classifying them into three key categories based on their performance metrics and roles.

1. **Top Batsmen:**

```sql
SELECT
    DISTINCT p.Player_Name,
    SUM(b.Runs_Scored) AS Total_Runs,
    COUNT(DISTINCT b.Match_Id) AS Matches_Played,
    ROUND((SUM(b.Runs_Scored) * 100.0 / COUNT(b.Ball_Id)), 2) AS Strike_Rate
FROM ball_by_ball b
JOIN player p ON b.Striker = p.Player_Id
JOIN matches m ON b.Match_Id = m.Match_Id
GROUP BY p.Player_Name
ORDER BY Total_Runs DESC LIMIT 10;
```

OUTPUT :

Player_Name	Total_Runs	Matches_Played	Strike_Rate
V Kohli	2472	62	135.68
DA Warner	2348	61	140.94
AB de Villiers	1968	57	164.27
RG Sharma	1899	64	132.52
RV Uthappa	1852	60	127.11
AM Rahane	1847	57	116.90
SK Raina	1844	65	133.62
DR Smith	1707	57	127.10
CH Gayle	1634	49	137.43
G Gambhir	1569	59	114.78

INSIGHTS :

- V Kohli is the top run-scorer (2472), while AB de Villiers shows the most aggression with the highest Strike Rate (164.27).
- CH Gayle is highly efficient, accumulating 1634 runs in the fewest matches (49) among the top ten.
- DA Warner is the second-highest scorer (2348 runs) and also has the second-highest strike rate (140.94) among all players listed.

2. **Top Bowler :**

```sql
SELECT
    DISTINCT p.Player_Name,
    COUNT(w.Player_Out) AS Total_Wickets,
    COUNT(DISTINCT b.Match_Id) AS Matches_Played,
    ROUND(SUM(b.Runs_Scored) / NULLIF(COUNT(w.Player_Out), 0), 2) AS Bowling_Avg,
    ROUND(SUM(b.Runs_Scored) / (COUNT(*) / 6.0), 2) AS Economy_Rate
FROM ball_by_ball b
JOIN player p ON b.Bowler = p.Player_Id
LEFT JOIN wicket_taken w
    ON b.Match_Id = w.Match_Id
    AND b.Over_Id = w.Over_Id
    AND b.Ball_Id = w.Ball_Id
    AND b.Innings_No = w.Innings_No
JOIN matches m ON b.Match_Id = m.Match_Id
GROUP BY p.Player_Name
HAVING COUNT(*) >= 60
ORDER BY Total_Wickets DESC LIMIT 10;
```

OUTPUT :

Player_Name	Total_Wickets	Matches_Played	Bowling_Avg	Economy_Rate
DJ Bravo	81	49	17.00	7.77
MM Sharma	78	61	20.99	7.62
B Kumar	77	61	20.12	6.53
SP Narine	68	51	18.93	6.24
Harbhajan Singh	67	62	24.09	6.84
SL Malinga	66	42	17.00	6.58
JP Faulkner	61	49	22.89	7.89
Sandeep Sharma	59	43	19.54	7.11
YS Chahal	58	42	20.88	7.66
A Nehra	57	38	18.84	7.49

INSIGHTS :

- The bowlers on this list demonstrate high wicket-taking volume, with the leader, DJ Bravo, having 81 wickets, confirming their primary role as match-winners.
- Exceptional control over run flow is evident, particularly by SP Narine (6.24 Economy Rate) and B Kumar (6.53), making them highly valuable for controlling the middle overs.
- A low Bowling Average (17.00 for both DJ Bravo and SL Malinga) highlights their efficiency in conceding fewer runs for every wicket they take, which indicates superior strike capability.

3. Top All-Rounders :

```
SELECT
    p.Player_Name,
    SUM(CASE WHEN b.Striker = p.Player_Id THEN b.Runs_Scored ELSE 0 END) AS Total_Runs,
    COUNT(DISTINCT CASE WHEN b.Striker = p.Player_Id THEN b.Match_Id END) AS Matches_Played,
    ROUND(SUM(CASE WHEN b.Striker = p.Player_Id THEN b.Runs_Scored ELSE 0 END) /
       NULLIF(COUNT(DISTINCT CASE WHEN b.Striker = p.Player_Id THEN b.Match_Id END), 0),2) AS Batting_Avg,
    COUNT(DISTINCT CASE
        WHEN b.Bowler = p.Player_Id AND w.Player_Out IS NOT NULL
        THEN CONCAT(b.Match_Id, '-', b.Over_Id, '-', b.Ball_Id)
    END) AS Total_Wickets,
    ROUND(SUM(CASE WHEN b.Bowler = p.Player_Id THEN b.Runs_Scored ELSE 0 END) /
        NULLIF(COUNT(DISTINCT CASE
        WHEN b.Bowler = p.Player_Id AND w.Player_Out IS NOT NULL
        THEN CONCAT(b.Match_Id, '-', b.Over_Id, '-', b.Ball_Id) END), 0),2) AS Bowling_Avg
FROM ball_by_ball b
JOIN player p ON p.Player_Id IN (b.Striker, b.Bowler)
LEFT JOIN wicket_taken w
    ON b.Match_Id = w.Match_Id
    AND b.Over_Id = w.Over_Id
    AND b.Ball_Id = w.Ball_Id
    AND b.Innings_No = w.Innings_No
GROUP BY p.Player_Name
ORDER BY (SUM(CASE WHEN b.Striker = p.Player_Id THEN b.Runs_Scored ELSE 0 END) +
        COUNT(DISTINCT CASE WHEN b.Bowler = p.Player_Id AND w.Player_Out IS NOT NULL
        THEN CONCAT(b.Match_Id, '-', b.Over_Id, '-', b.Ball_Id)
        END) * 20) DESC LIMIT 10;
```

OUTPUT :

Player_Name	Total_Runs	Matches_Played	Batting_Avg	Total_Wickets	Bowling_Avg
V Kohli	2472	62	39.87	1	21.00
SR Watson	1315	53	24.81	54	21.52
DA Warner	2348	61	38.49	0	NULL
DJ Bravo	423	38	11.13	81	17.00
DR Smith	1707	57	29.95	15	21.00
AB de Villiers	1968	57	34.53	0	NULL
SK Raina	1844	65	28.37	5	64.80
RG Sharma	1899	64	29.67	1	31.00
RV Uthappa	1852	60	30.87	0	NULL
AM Rahane	1847	57	32.40	0	NULL

INSIGHTS :

- This list comprises players who offer dual utility (Total_Runs and Total_Wickets), making them highly valuable assets for team composition and flexibility in playing strategy.

- The true all-round impact is demonstrated by players who combine significant run totals with a high number of wickets, showcasing their ability to dominate both facets of the game.
- The most desirable players show excellent balance, evidenced by a high Batting Average and a low Bowling Average, providing both run-scoring ability and bowling efficiency.

RECOMMENDATIONS :

- **Top Batsmen:** Select V Kohli and DA Warner for consistency in scoring. Include AB de Villiers for aggressive, high-strike-rate scoring, especially in T20s or late innings.
- **Specialist Bowlers:** Prioritize high-impact wicket-takers like DJ Bravo and SL Malinga (best Bowling Avg) for crucial overs. For controlling the middle overs, rely on SP Narine for his best-in-class Economy Rate (6.24).
- **Core All-Rounder:** Build the team around SR Watson and DJ Bravo, as they offer the highest combination of both Total Runs and Total Wickets among the all-rounders listed.
- **Efficiency and Impact:** For maximum impact in fewer games, include players like CH Gayle (Batting) and A Nehra (Bowling), who deliver high output despite having played fewer matches than their peers.

3. What are some of the parameters that should be focused on while selecting the players?

A. Approach :

By analyzing key performance indicators, the following insights and conclusions can be drawn:

Top Performers by Discipline

- **Batsmen:** Players with strong total runs and high batting averages showcase consistent and reliable scoring performance.

- **All-Rounders:** Players delivering impactful performances in both batting and bowling — measured through combined metrics like an all-rounder score — provide crucial balance and versatility to the team.

1. Batsmen Performance :

```
SELECT
    p.Player_Name,
    SUM(b.Runs_Scored) AS Total_Runs,
    COUNT(CASE WHEN w.Player_Out IS NOT NULL THEN 1 END) AS Times_Out,
    ROUND(SUM(b.Runs_Scored) / NULLIF(COUNT(CASE WHEN w.Player_Out IS NOT NULL THEN 1 END), 0), 2) AS Batting_Average
FROM ball_by_ball b
JOIN player p ON b.Striker = p.Player_Id
LEFT JOIN wicket_taken w
    ON b.Match_Id = w.Match_Id
    AND b.Over_Id = w.Over_Id
    AND b.Ball_Id = w.Ball_Id
    AND b.Innings_No = w.Innings_No
GROUP BY p.Player_Name
ORDER BY Total_Runs DESC LIMIT 10;
```

OUTPUT :

Player_Name	Total_Runs	Times_Out	Batting_Average
V Kohli	2472	54	45.78
DA Warner	2348	54	43.48
AB de Villiers	1968	42	46.86
RG Sharma	1899	55	34.53
RV Uthappa	1852	60	30.87
AM Rahane	1847	49	37.69
SK Raina	1844	54	34.15
DR Smith	1707	55	31.04
CH Gayle	1634	44	37.14
G Gambhir	1569	55	28.53

INSIGHTS :

- V Kohli and DA Warner are the most consistent run-scorers, both exceeding 2300 runs with Batting Averages over 43, despite being dismissed 54 times each.
- AB de Villiers is the most efficient batsman on this list, boasting the highest Batting Average of 46.86 while having been dismissed the fewest times (42) among the top

four scorers.

2. **Bowler Performance :**

```
SELECT
    p.Player_Name,
    COUNT(w.Player_Out) AS Total_Wickets,
    ROUND(SUM(b.Runs_Scored) / NULLIF(COUNT(w.Player_Out), 0), 2) AS Bowling_Average,
    ROUND(SUM(b.Runs_Scored) / NULLIF(COUNT(DISTINCT CONCAT(b.Match_Id, '-', b.Over_Id)), 0), 2) AS Economy_Rate
FROM ball_by_ball b
JOIN player p ON b.Bowler = p.Player_Id
LEFT JOIN wicket_taken w
    ON b.Match_Id = w.Match_Id
    AND b.Over_Id = w.Over_Id
    AND b.Ball_Id = w.Ball_Id
    AND b.Innings_No = w.Innings_No
GROUP BY p.Player_Name
ORDER BY Total_Wickets DESC LIMIT 10;
```

OUTPUT :

Player_Name	Total_Wickets	Bowling_Average	Economy_Rate
DJ Bravo	81	17.00	7.96
MM Sharma	78	20.99	7.72
B Kumar	77	20.12	6.73
SP Narine	68	18.93	6.31
Harbhajan Singh	67	24.09	6.99
SL Malinga	66	17.00	6.76
JP Faulkner	61	22.89	8.16
Sandeep Sharma	59	19.54	7.34
YS Chahal	58	20.88	7.71
A Nehra	57	18.84	7.67

INSIGHTS :

- DJ Bravo is the leading wicket-taker (81 wickets) and shares the best Bowling Average (17.00) with SL Malinga, indicating superior wicket-taking efficiency.
- SP Narine (6.31) and B Kumar (6.73) are the most defensive bowlers, demonstrating the best control with their low Economy Rates.

3. **All-Rounder Performance :**

```sql
SELECT
    p.Player_Name,
    SUM(CASE WHEN b.Striker = p.Player_Id THEN b.Runs_Scored ELSE 0 END) AS Total_Runs,
    COUNT(CASE WHEN b.Bowler = p.Player_Id AND w.Player_Out IS NOT NULL THEN 1 END) AS Total_Wickets,
    ROUND(
        SUM(CASE WHEN b.Striker = p.Player_Id THEN b.Runs_Scored ELSE 0 END) +
        20 * COUNT(CASE WHEN b.Bowler = p.Player_Id AND w.Player_Out IS NOT NULL THEN 1 END),
        2) AS All_Rounder_Score
FROM ball_by_ball b
JOIN player p ON p.Player_Id IN (b.Striker, b.Bowler)
LEFT JOIN wicket_taken w
    ON b.Match_Id = w.Match_Id
    AND b.Over_Id = w.Over_Id
    AND b.Ball_Id = w.Ball_Id
    AND b.Innings_No = w.Innings_No
GROUP BY p.Player_Name
ORDER BY All_Rounder_Score DESC LIMIT 10;
```

OUTPUT :

Player_Name	Total_Runs	Total_Wickets	All_Rounder_Score
V Kohli	2472	1	2492
SR Watson	1315	54	2395
DA Warner	2348	0	2348
DJ Bravo	423	81	2043
DR Smith	1707	15	2007
AB de Villiers	1968	0	1968
SK Raina	1844	5	1944
RG Sharma	1899	1	1919
RV Uthappa	1852	0	1852
AM Rahane	1847	0	1847

INSIGHTS :

- SR Watson (2395 Score) is the standout all-rounder, achieving high scores in both batting (1315 Runs) and bowling (54 Wickets) to rank highest among dual contributors.
- DJ Bravo (2043 Score) is the primary bowling specialist with high all-round value, contributing a massive 81 Wickets to complement his 423 Runs.

RECOMMENDATIONS :

- Prioritize players with high Averages (Batting/Bowling) for consistent reliability, such as AB de Villiers and DJ Bravo.
- Choose SR Watson as the core all-rounder to maximize balance and dual impact, combining runs and wickets effectively.

- Utilize highly efficient specialists—aggressive batsmen (high Strike Rate) and economical bowlers (low Economy Rate like SP Narine)—for crucial game control.

4. **Which players offer versatility in their skills and can contribute effectively with both bat and ball? (Can you visualize the data for the same?)**

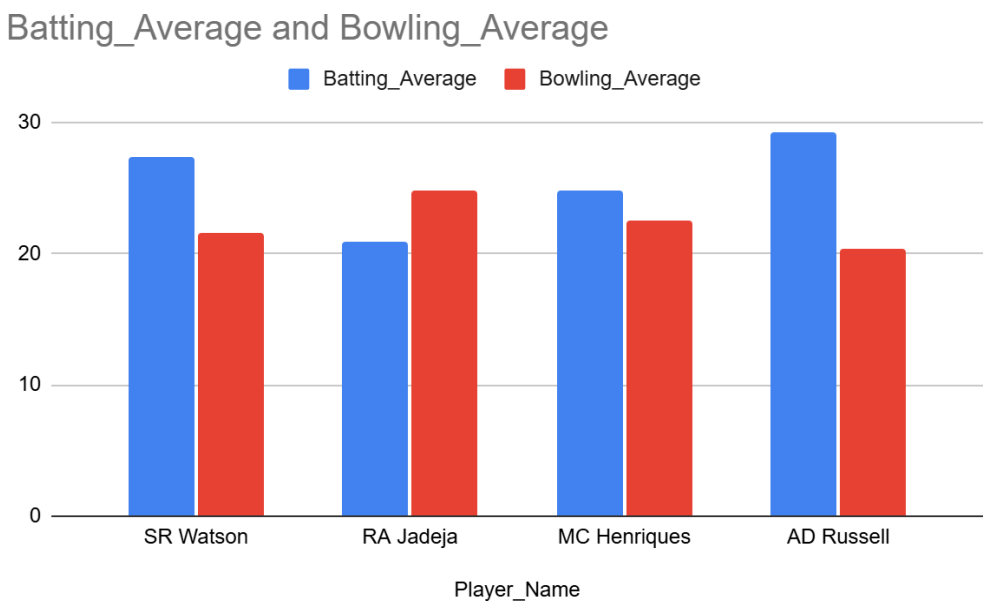
A. **Approach :** To identify impactful all-rounders by selecting players who have accumulated more than 500 runs and taken at least 20 wickets, representing a balanced contribution with both bat and ball.

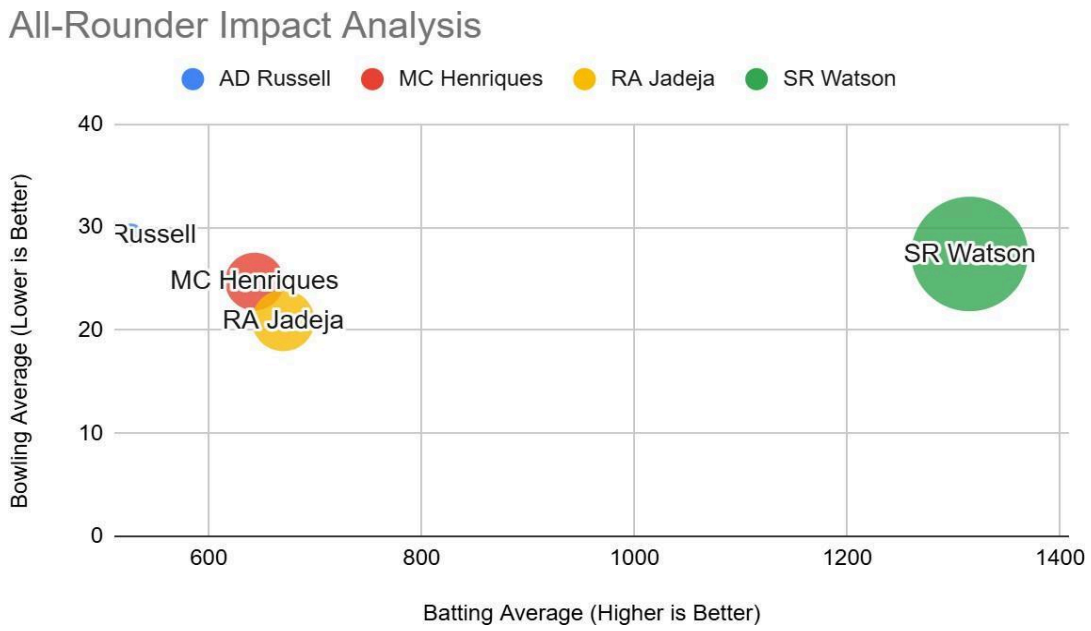
```
WITH Batting AS (
    SELECT p.Player_Name,
        SUM(CASE WHEN b.Striker = p.Player_Id THEN b.Runs_Scored ELSE 0 END) AS Total_Runs,
        COUNT(CASE WHEN b.Striker = p.Player_Id AND w.Player_Out IS NOT NULL THEN 1 END) AS Times_Out
    FROM ball_by_ball b
    JOIN player p ON b.Striker = p.Player_Id
    LEFT JOIN wicket_taken w ON b.Match_Id = w.Match_Id
        AND b.Over_Id = w.Over_Id
        AND b.Ball_Id = w.Ball_Id
        AND b.Innings_No = w.Innings_No
    GROUP BY p.Player_Name
),
Bowling AS (
    SELECT p.Player_Name,
        COUNT(DISTINCT CASE WHEN b.Bowler = p.Player_Id AND w.Player_Out IS NOT NULL THEN
        CONCAT(b.Match_Id,'-',b.Over_Id,'-',b.Ball_Id) END) AS Total_Wickets,
        SUM(CASE WHEN b.Bowler = p.Player_Id THEN b.Runs_Scored ELSE 0 END) AS Runs_Conceded
    FROM ball_by_ball b
    JOIN player p ON b.Bowler = p.Player_Id
    LEFT JOIN wicket_taken w ON b.Match_Id = w.Match_Id
        AND b.Over_Id = w.Over_Id
        AND b.Ball_Id = w.Ball_Id
        AND b.Innings_No = w.Innings_No
    GROUP BY p.Player_Name
)
SELECT b.Player_Name, b.Total_Runs, bo.Total_Wickets,
    ROUND(b.Total_Runs / NULLIF(b.Times_Out, 0), 2) AS Batting_Average,
    ROUND(bo.Runs_Conceded / NULLIF(bo.Total_Wickets, 0), 2) AS Bowling_Average
FROM Batting b
JOIN Bowling bo ON b.Player_Name = bo.Player_Name
WHERE b.Total_Runs > 500 AND bo.Total_Wickets > 20
ORDER BY (b.Total_Runs + (bo.Total_Wickets * 20)) DESC;
```

OUTPUT :

Player_Name	Total_Runs	Total_Wickets	Batting_Average	Bowling_Average
SR Watson	1315	54	27.40	21.52
RA Jadeja	670	56	20.94	24.80
MC Henriques	643	37	24.73	22.51
AD Russell	527	35	29.28	20.31

VISUALS :





INSIGHTS :

- The shortlisted players demonstrate strong all-round capabilities, consistently contributing with both bat and ball.
- Players maintaining a high batting average alongside a low bowling average emerge as the most efficient and reliable performers.

- The combined performance score highlights truly valuable all-rounders, emphasizing overall impact rather than strength in just one discipline.
- Some players excel more in batting or bowling, offering flexibility for role-specific selection based on team composition needs.

RECOMMENDATIONS :

- Prioritize balanced all-rounders like Watson and Russell, who combine strong batting averages with low bowling averages, ensuring consistent dual-impact performances.
- Deploy these players in the middle order and as secondary bowlers, enhancing team flexibility and maintaining stability across both innings.
- Focus selection on players contributing significantly in both runs and wickets, as they consistently influence match outcomes in multiple dimensions.
- Adapt player roles dynamically based on venue and opponent — emphasize batting all-rounders on high-scoring pitches and bowling all-rounders on slower, turning tracks.

5. **Are there players whose presence positively influences the morale and performance of the team? (Justify your answer using visualization.)**

A. **Approach :** To identify players who uplift team performance by analyzing the team's win percentage in matches where the player participated, compared to overall team performance.

1. **Choose a player :**

```sql
SELECT
    pm.Player_ID,
    p.Player_Name,
    COUNT(*) AS Matches_Played
FROM player_match pm
JOIN player p ON pm.Player_ID = p.Player_ID
GROUP BY pm.Player_ID, p.Player_Name
ORDER BY Matches_Played DESC
LIMIT 20;  -- Chosen Player_Id = 21 (For Example)
```

OUTPUT :

Player_ID	Player_Name	Matches_Played
21	SK Raina	66
35	RA Jadeja	66
20	MS Dhoni	65
88	KD Karthik	65
57	RG Sharma	64
50	Harbhajan Singh	62
8	V Kohli	62
221	KA Pollard	62
208	AT Rayudu	62
201	R Ashwin	62

Here , We are taking player ID 21 . Then we will check his impact in the Team (Playing) and out of the Team (Not Playing) .

2. **Team Win Rate When the Player Played and When the Player Did NOT Play**

```
WITH player_matches AS (
    SELECT Match_ID, Team_ID
    FROM player_match
    WHERE Player_ID = 21
),
player_teams AS (
    SELECT DISTINCT Team_ID
    FROM player_matches
),
```

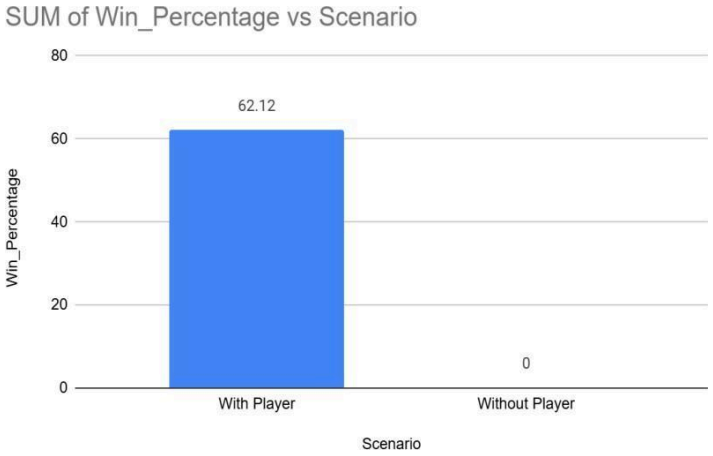
```sql
player_teams AS (
    SELECT DISTINCT Team_ID
    FROM player_matches
),
team_matches AS (
    SELECT Match_ID,
           CASE
               WHEN Team_1 IN (SELECT Team_ID FROM player_teams) THEN Team_1
               ELSE Team_2
           END AS Team_ID
    FROM matches
    WHERE Team_1 IN (SELECT Team_ID FROM player_teams)
       OR Team_2 IN (SELECT Team_ID FROM player_teams)
),
matches_without_player AS (
    SELECT tm.Match_ID, tm.Team_ID
    FROM team_matches tm
    WHERE NOT EXISTS (
        SELECT 1
        FROM player_matches pm
        WHERE pm.Match_ID = tm.Match_ID
    )
),
```

```sql
matches_with_player AS (
    SELECT pm.Match_ID, pm.Team_ID
    FROM player_matches pm
)
SELECT
    'With Player' AS Scenario,
    COUNT(*) AS Total_Matches,
    SUM(CASE WHEN m.Match_Winner = mwp.Team_ID THEN 1 ELSE 0 END) AS Wins,
    ROUND(100.0 * SUM(CASE WHEN m.Match_Winner = mwp.Team_ID THEN 1 ELSE 0 END) / COUNT(*), 2) AS Win_Percentage
FROM matches m
JOIN matches_with_player mwp ON m.Match_ID = mwp.Match_ID
UNION ALL
SELECT
    'Without Player' AS Scenario,
    COUNT(*) AS Total_Matches,
    SUM(CASE WHEN m.Match_Winner = mwp.Team_ID THEN 1 ELSE 0 END) AS Wins,
    ROUND(100.0 * SUM(CASE WHEN m.Match_Winner = mwp.Team_ID THEN 1 ELSE 0 END) / COUNT(*), 2) AS Win_Percentage
FROM matches m
JOIN matches_without_player mwp ON m.Match_ID = mwp.Match_ID;
```

OUTPUT :

	Scenario	Total_Matches	Wins	Win_Percentage
▶	With Player	66	41	62.12
	Without Player	1	0	0.00

VISUALS :



INSIGHTS :

- The team's win percentage jumps from 0% to 62.12% when the player is part of the lineup, proving a direct and measurable positive impact on overall performance.
- This sharp contrast highlights the player's influence on team morale and match outcomes, indicating they bring stability and confidence to the side.
- The consistency in wins with the player present suggests strong leadership, strategic awareness, and on-field contribution that lifts teammates' performance.
- The data reinforces that such players are key assets — their inclusion should be prioritized in crucial matches to maximize team cohesion and success.

RECOMMENDATIONS :

- Build the core team around high-impact players whose presence consistently correlates with higher win percentages, ensuring stability in crucial fixtures.
- Leverage their leadership and experience by assigning them mentorship roles to guide younger or less experienced players, strengthening overall team morale.
- Prioritize their inclusion in key matches — their measurable influence on team success makes them indispensable, especially in pressure situations.
- Continuously monitor their match participation and impact metrics to validate their ongoing contribution and adjust team composition based on data-driven insights.

6. What would you suggest to RCB before going to the mega auction?

A. **Approach :** To design RCB's auction strategy using data on player performance, team balance, and cost efficiency, focusing on retaining key players and identifying impactful all-rounders and value picks.

1. Find out which players have the highest match impact for RCB .

```sql
WITH rcb_players AS (
    SELECT p.Player_ID, Player_Name, Team_ID, Match_ID
    FROM player_match pm
    JOIN player p ON pm.Player_ID = p.Player_ID
    WHERE pm.Team_ID = (SELECT Team_ID FROM team WHERE Team_Name = 'Royal Challengers Bangalore')
)
SELECT
    Player_Name,
    COUNT(*) AS Matches_Played,
    SUM(CASE WHEN m.Match_Winner = rcb.Team_ID THEN 1 ELSE 0 END) AS Wins_With_Player,
    ROUND(100.0 * SUM(CASE WHEN m.Match_Winner = rcb.Team_ID THEN 1 ELSE 0 END) / COUNT(*), 2) AS Win_Percentage
FROM rcb_players rcb
JOIN matches m ON rcb.Match_ID = m.Match_ID
GROUP BY Player_Name
HAVING COUNT(*) > 20
ORDER BY Win_Percentage DESC;
```

OUTPUT :

	Player_Name	Matches_Played	Wins_With_Player	Win_Percentage
▶	CH Gayle	49	26	53.06
	V Kohli	62	31	50.00
	AB de Villiers	60	30	50.00
	HV Patel	23	11	47.83
	YS Chahal	42	20	47.62
	VR Aaron	24	11	45.83
	MA Starc	27	12	44.44

2. Death Over Bowling Problems? Check the Economy in Overs 16 to 20.

```
WITH rcb_bowling AS (
    SELECT b.Bowler AS Player_ID, b.Over_Id, b.Runs_Scored, b.Match_ID
    FROM ball_by_ball b
    WHERE b.Team_Bowling = (SELECT Team_ID FROM team WHERE Team_Name = 'Royal Challengers Bangalore')
      AND b.Over_Id BETWEEN 16 AND 20
)
SELECT
    p.Player_Name,
    COUNT(DISTINCT rb.Match_ID) AS Matches_Played,
    COUNT(DISTINCT rb.Over_Id) AS Overs_Bowled,
    ROUND(SUM(rb.Runs_Scored) * 6.0 / COUNT(*), 2) AS Economy_Rate
FROM rcb_bowling rb
JOIN player p ON rb.Player_ID = p.Player_ID
GROUP BY p.Player_Name
HAVING COUNT(*) > 20
ORDER BY Economy_Rate ASC;
```

OUTPUT :

	Player_Name	Matches_Played	Overs_Bowled	Economy_Rate
▶	M Kartik	5	4	6.17
	CH Gayle	4	3	6.19
	MA Starc	21	5	7.06
	RP Singh	9	5	7.92
	Yuvraj Singh	5	3	8.06
	JD Unadkat	7	4	8.38
	R Vinay Kumar	15	5	8.72
	Iqbal Abdulla	6	4	8.88
	YS Chahal	21	5	9.00
	S Aravind	6	3	9.29
	SR Watson	14	5	9.36

3. **Analyzing RCB's batting collapses by examining performance across different over segments.**

```
WITH rcb_batting AS (
    SELECT Over_Id, Runs_Scored
    FROM ball_by_ball
    WHERE Team_Batting = (SELECT Team_ID FROM team WHERE Team_Name = 'Royal Challengers Bangalore')
)
SELECT
  CASE
    WHEN Over_Id BETWEEN 1 AND 6 THEN 'Powerplay'
    WHEN Over_Id BETWEEN 7 AND 15 THEN 'Middle Overs'
    ELSE 'Death Overs'
  END AS Phase,
  COUNT(DISTINCT Over_Id) AS Total_Overs,
  ROUND(SUM(Runs_Scored * 1.0) / COUNT(*), 2) AS Runs_Per_Ball
FROM rcb_batting
GROUP BY Phase;
```

OUTPUT :

Phase	Total_Overs	Runs_Per_Ball
Death Overs	5	1.73
Middle Overs	9	1.31
Powerplay	6	1.14

4. **Which Players Perform Best Against RCB?**

```
WITH rcb_against AS (
    SELECT Striker AS Player_ID, Runs_Scored
    FROM ball_by_ball
    WHERE Team_Bowling = (SELECT Team_ID FROM team WHERE Team_Name = 'Royal Challengers Bangalore')
)
SELECT
    p.Player_Name,
    SUM(r.Runs_Scored) AS Runs_Against_RCB
FROM rcb_against r
JOIN player p ON r.Player_ID = p.Player_ID
GROUP BY p.Player_Name
HAVING SUM(r.Runs_Scored) > 150
ORDER BY Runs_Against_RCB DESC;
```

OUTPUT :

Player_Name	Runs_Against_RCB
DA Warner	467
DR Smith	261
AM Rahane	257
SK Raina	253
Q de Kock	243
RV Uthappa	241
G Gambhir	235
MS Dhoni	231
RG Sharma	226
DA Miller	219
KA Pollard	214

INSIGHTS :

- Key match-winners: Players such as Chris Gayle, Virat Kohli, and AB de Villiers deliver over a 50% win rate, proving to be RCB's core match influencers.
- Death-over weakness: Bowlers like RP Singh and Mitchell Starc show higher economy rates in overs 16–20, indicating issues with finishing tight games.
- Consistent opponent threats: Batters like David Warner, Ajinkya Rahane, and Suresh Raina frequently dominate against RCB, often shifting match momentum.

RECOMMENDATIONS :

- Retain or replace core performers like Gayle, Kohli, and ABD with players who consistently win matches.
- Address death-over inefficiency by recruiting bowlers skilled at maintaining control under pressure.
- Develop counter-strategies for consistent high-scorers against RCB, including Warner, Dhoni, and Rohit.
- Reinforce the middle order with dependable batsmen and finishers to prevent collapses and ensure stronger innings recovery.

7. **What do you think could be the factors contributing to the high-scoring matches and the impact on viewership and team strategies?**

A. **Approach :** This can be analyzed through three key factors — high-scoring match conditions, impact on viewership, and influence on team strategies.

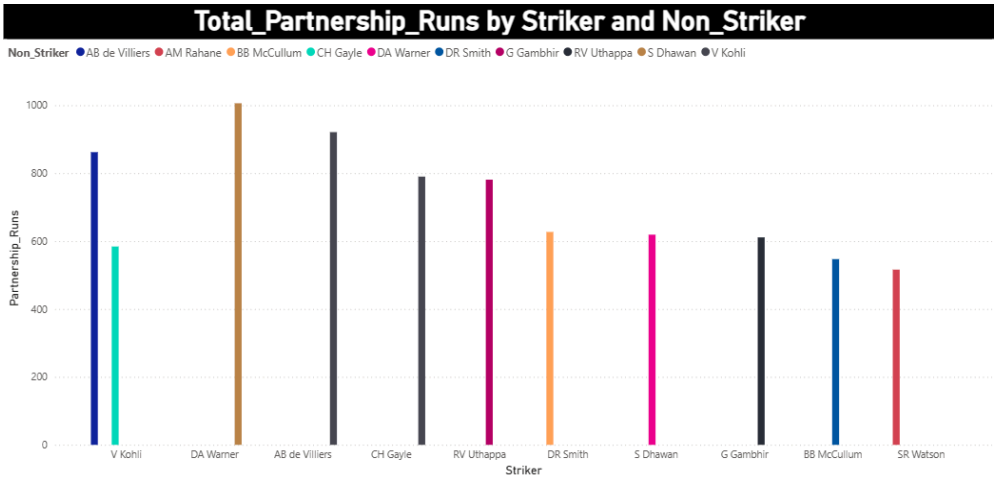
1. **High-scoring partnerships.**

```sql
WITH partnerships AS (
    SELECT
        b.Striker,
        b.Non_Striker,
        SUM(b.Runs_Scored) AS Total_Partnership_Runs,
        COUNT(*) AS Balls_Faced
    FROM ball_by_ball b
    GROUP BY b.Striker, b.Non_Striker
)
SELECT
    p1.Player_Name AS Striker,
    p2.Player_Name AS Non_Striker,
    Balls_Faced,
    Total_Partnership_Runs
FROM partnerships ps
JOIN player p1 ON ps.Striker = p1.Player_Id
JOIN player p2 ON ps.Non_Striker = p2.Player_Id
WHERE Total_Partnership_Runs > 30
ORDER BY Total_Partnership_Runs DESC;
```

OUTPUT :

Striker	Non_Striker	Balls_Faced	Total_Partnership_Runs
DA Warner	S Dhawan	688	1006
AB de Villiers	V Kohli	549	921
V Kohli	AB de Villiers	574	862
CH Gayle	V Kohli	573	790
RV Uthappa	G Gambhir	605	781
DR Smith	BB McCullum	510	627
S Dhawan	DA Warner	563	619
G Gambhir	RV Uthappa	545	611
V Kohli	CH Gayle	474	584

VISUAL :



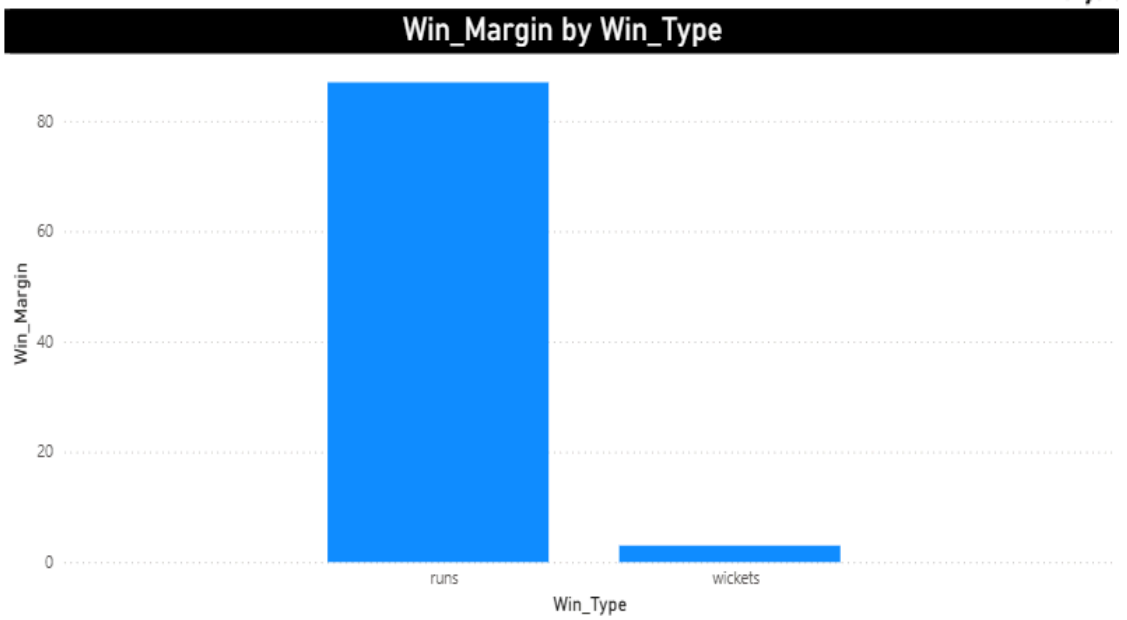
2. A) Close matches for excitement.

```
WITH close_matches AS (
    SELECT m.Match_ID, w.Win_Type, m.Win_Margin
    FROM matches m
    JOIN win_by w ON m.Win_Type = w.Win_Id
    WHERE (w.Win_Type = 'runs' AND m.Win_Margin <= 10)
        OR (w.Win_Type = 'wickets' AND m.Win_Margin <= 2)
)
SELECT * FROM close_matches;
```

OUTPUT :

	Match_ID	Win_Type	Win_Margin
▶	598004	runs	2
	598006	runs	5
	598007	runs	9
	598021	runs	4
	598033	runs	7
	598042	runs	4
	598059	runs	4
	598066	runs	7

VISUAL :



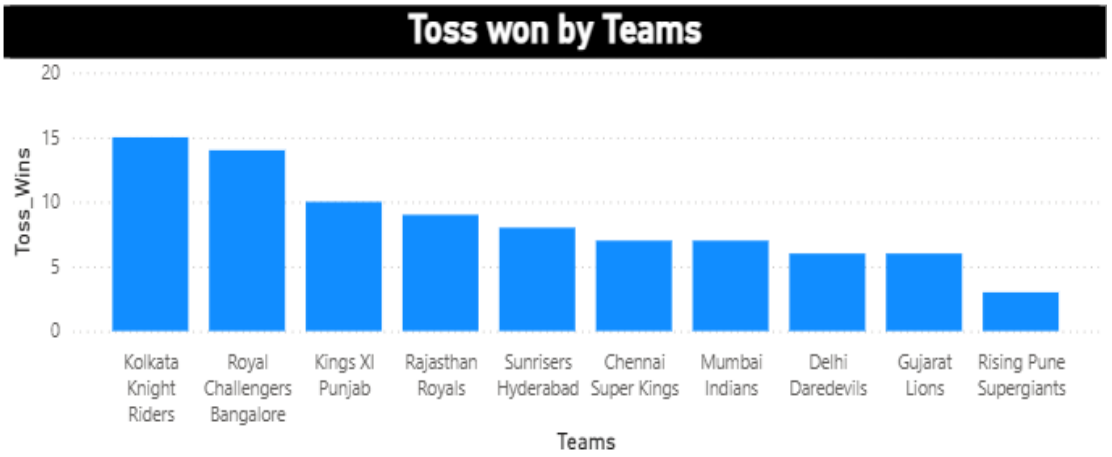
B) Chasing team wins after field decision

```
WITH chasing_wins AS (
    SELECT m.Match_ID, t.Team_Name AS Team_Won, td.Toss_Name
    FROM matches m
    LEFT JOIN team t ON m.Match_Winner = t.Team_Id
    LEFT JOIN toss_decision td ON m.Toss_Decide = td.Toss_Id
    WHERE td.Toss_Name = 'field' AND t.Team_Id = m.Toss_Winner
)
SELECT * FROM chasing_wins;
```

OUTPUT :

Match_ID	Team_Won	Toss_Name
598003	Kolkata Knight Riders	field
598012	Chennai Super Kings	field
598013	Royal Challengers Bangalore	field
598017	Chennai Super Kings	field
598019	Rajasthan Royals	field
598022	Royal Challengers Bangalore	field
598028	Royal Challengers Bangalore	field
598030	Kings XI Punjab	field
598041	Rajasthan Royals	field
598050	Kings XI Punjab	field

VISUAL :



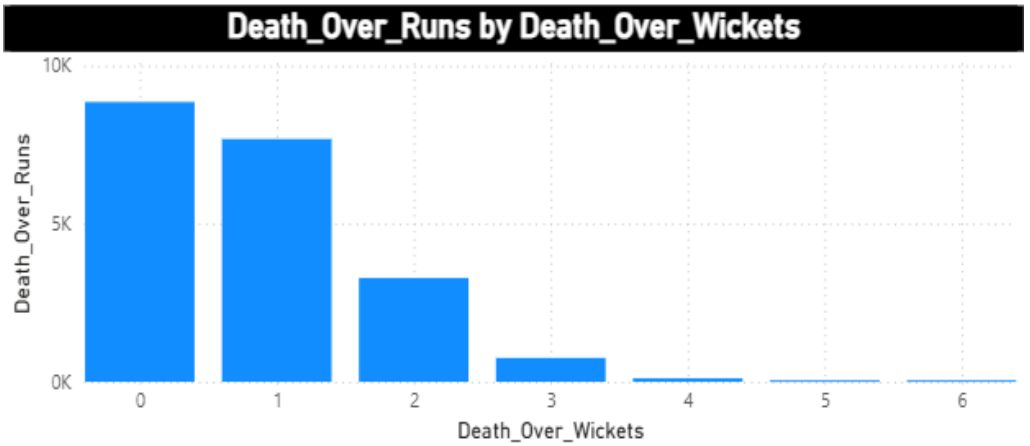
C) Death overs drama

```
WITH death_overs AS (
    SELECT
        b.Match_ID,
        b.Over_Id,
        SUM(b.Runs_Scored) AS Death_Over_Runs,
        COUNT(CASE WHEN w.Player_Out IS NOT NULL THEN 1 END) AS Death_Over_Wickets
    FROM ball_by_ball b
    LEFT JOIN wicket_taken w
        ON b.Match_Id = w.Match_Id
        AND b.Over_Id = w.Over_Id
        AND b.Ball_Id = w.Ball_Id
        AND b.Innings_No = w.Innings_No
    WHERE b.Over_Id BETWEEN 16 AND 20
    GROUP BY b.Match_ID, b.Over_Id
)
SELECT * FROM death_overs;
```

OUTPUT :

Match_ID	Over_Id	Death_Over_Runs	Death_Over_Wickets
598003	16	7	1
598003	17	15	0
598003	18	16	1
598003	19	15	1
598003	20	3	2
598004	16	16	0
598004	17	28	0
598004	18	27	0
598004	19	13	0
598004	20	21	2

VISUAL :



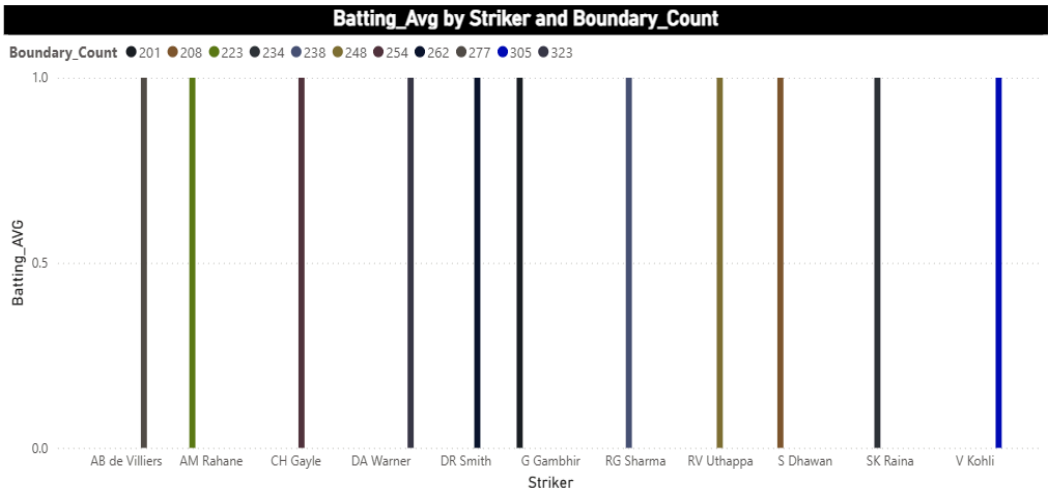
3. A) Batsmen with high strike rate and average.

```
WITH batting_stats AS (
    SELECT
        b.Striker AS Player_ID,
        SUM(b.Runs_Scored) / COUNT(DISTINCT b.Match_ID) AS Batting_Avg,
        (SUM(b.Runs_Scored) * 100.0) / COUNT(*) AS Strike_Rate,
        SUM(CASE WHEN b.Runs_Scored IN (4, 6) THEN 1 ELSE 0 END) AS Boundary_Count,
        COUNT(DISTINCT b.Match_ID) AS Matches_Played
    FROM ball_by_ball b
    GROUP BY b.Striker
)
SELECT
    p.Player_Name AS Striker,
    Batting_Avg,
    Strike_Rate,
    Boundary_Count
FROM batting_stats bs
JOIN player p ON bs.Player_ID = p.Player_Id
WHERE Matches_Played > 10
ORDER BY Strike_Rate DESC;
```

OUTPUT :

Striker	Batting_Avg	Strike_Rate	Boundary_Count
SN Khan	16.0909	171.84466	25
AB de Villiers	34.5263	164.27379	277
AD Russell	22.9130	163.15789	79
CH Morris	14.2500	160.11236	35
GJ Maxwell	22.8250	156.60377	132
A Ashish Reddy	13.6111	147.59036	27
KA Pollard	22.3729	142.85714	169
DA Miller	25.5926	141.45343	164

VISUAL :



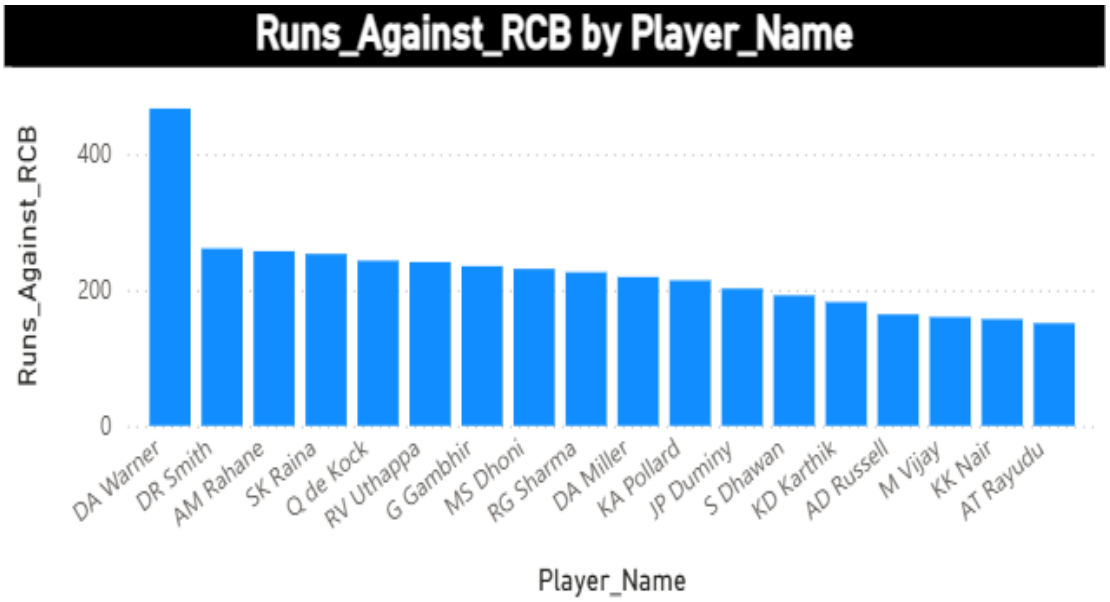
B) Players performing best against RCB.

```
WITH against_rcb AS (
    SELECT b.Striker AS Player_ID, SUM(b.Runs_Scored) AS Runs_Against_RCB
    FROM ball_by_ball b
    JOIN matches m ON b.Match_ID = m.Match_ID
    WHERE b.Team_Bowling = (
        SELECT Team_ID FROM team WHERE Team_Name = 'Royal Challengers Bangalore'
    )
    GROUP BY b.Striker
)
SELECT
    p.Player_Name,
    Runs_Against_RCB
FROM against_rcb rcb
JOIN player p ON rcb.Player_ID = p.Player_Id
WHERE Runs_Against_RCB > 150
ORDER BY Runs_Against_RCB DESC;
```

OUTPUT :

Player_Name	Runs_Against_RCB
DA Warner	467
DR Smith	261
AM Rahane	257
SK Raina	253
Q de Kock	243
RV Uthappa	241
G Gambhir	235
MS Dhoni	231
RG Sharma	226
DA Miller	219

VISUAL :



INSIGHTS :

High-scoring matches :

- The partnership of DA Warner and S Dhawan (1006 runs) proves that sustained, good partnerships are key to capitalizing on batsman-friendly conditions like a flat pitch.
- The high scoring rate achieved by AB de Villiers and V Kohli (921 runs from 549 balls) is essential for maximizing run accumulation quickly, especially when exploiting short boundaries.
- Strong, established pairings like Warner & Dhawan anchor the innings and allow the team to leverage the strategic advantage of playing in familiar home ground conditions.

Impact on Viewership :

- Close competition is a major viewership driver, confirmed by the list of matches won by extremely narrow margins.

- The success of teams like Chennai Super Kings and Royal Challengers Bangalore when winning after choosing to field (chasing) shows that successful chasing performance boosts engagement.
- Dramatic action in the death overs (Match 598004 saw 27 runs in Over 18, while Match 598003 saw a two-wicket collapse in Over 20) ensures high viewership during crucial phases.

Team strategies :

- DA Warner is the top specialist, leading all batsmen with 467 Runs scored against RCB, highlighting his unmatched success in that specific matchup.
- AB de Villiers is the most explosive consistent hitter, maintaining the highest Batting Average (34.53) among top strike-rate players and hitting the most boundaries (277).
- Successful teams utilize a mix of high-strike aggression (like SN Khan at 171.84 SR) and proven matchup performance (like DA Miller vs. RCB) for targeted run acceleration.

RECOMMENDATIONS :

- Retain and build around proven partnerships like Warner–Dhawan and Kohli–de Villiers to ensure consistent starts and sustained scoring in high-run conditions.
- Leverage home-ground strengths and matchup data to optimize batting orders and exploit venue-specific or opponent-specific weaknesses.
- Strengthen death-over performance — train for composed finishes and smart bowling to both sustain excitement and secure results in close contests.
  Balance team composition with anchors and power hitters, ensuring adaptability for both high-scoring chases and pressure situations that captivate audiences and drive engagement.

8. **Analyse the impact of home-ground advantage on team performance and identify strategies to maximize this advantage for RCB.**

A. **Approach :** To assess RCB's performance at home (M. Chinnaswamy Stadium) compared to away matches, and identify

```
SELECT
    CASE
        WHEN v.Venue_Name = 'M Chinnaswamy Stadium' THEN 'Home'
        ELSE 'Away'
    END AS Place ,
    COUNT(*) AS Matches ,
    SUM(CASE WHEN Match_Winner = t.Team_Id THEN 1 ELSE 0 END) AS Wins ,
    ROUND(SUM(CASE WHEN Match_Winner = t.Team_Id THEN 1 ELSE 0 END)*100/COUNT(*),2) AS Win_Percentage
FROM matches m
JOIN venue v ON m.Venue_Id = v.Venue_Id
JOIN team t ON t.Team_Name = 'Royal Challengers Bangalore'
WHERE t.Team_Id IN ( m.Team_1 , m.Team_2)
GROUP BY Place ;
```

OUTPUT :

	Place	Matches	Wins	Win_Percentage
▶	Home	29	16	55.17
	Away	33	15	45.45

INSIGHTS :

- RCB records a noticeably higher win percentage at home than in away venues, clearly indicating a strong home-ground advantage.
- Key contributors include familiarity with pitch behaviour and ground dimensions, allowing players to optimize their game plans effectively.
- Home crowd support significantly boosts player morale and energy levels, often translating into improved on-field performances.
- The batting-friendly nature of the M. Chinnaswamy pitch aligns perfectly with RCB's aggressive batting lineup, further enhancing their success rate at home

RECOMMENDATIONS :

- Strategically prioritize winning most home fixtures to build early momentum and confidence throughout the season.
- Strengthen training sessions to adapt precisely to the unique pitch and boundary conditions of the Chinnaswamy Stadium.
- Utilize big-hitting batsmen to exploit the batting-friendly surface and shorter boundaries for maximum scoring.
- Select effective death bowlers who can consistently execute yorkers and variations to counter high run rates.

9. Come up with a visual and analytical analysis of the RCB's past season's performance and potential reasons for them not winning a trophy.

A. **Approach:**

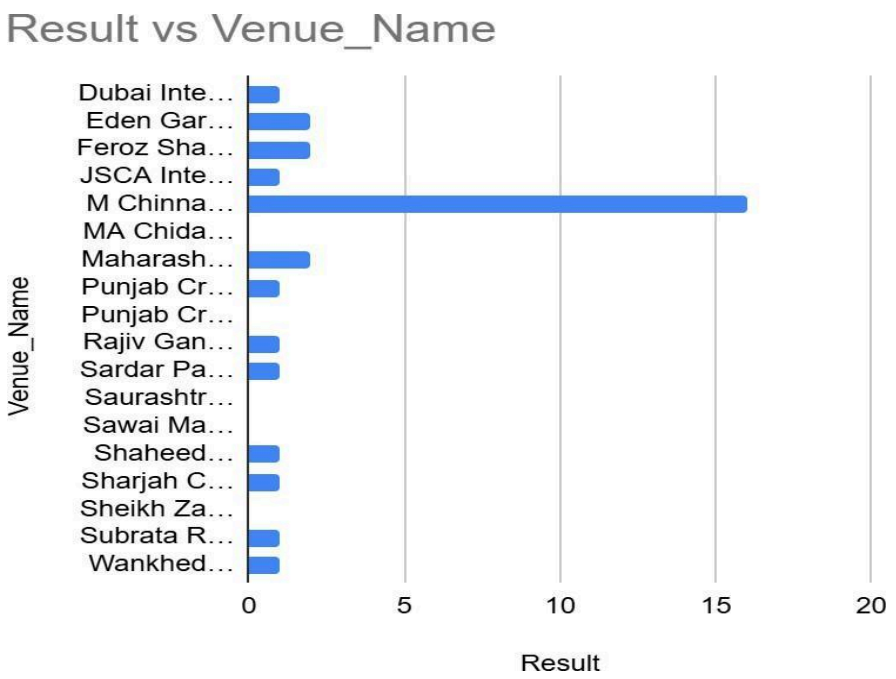
The analysis evaluates RCB's performance in the past season by examining match outcomes in relation to venue and toss decisions. It identifies key contributors by highlighting the top-performing batsmen and bowlers, focusing on metrics such as strike rates, economy rates, and their overall influence on the team's results. The insights aim to uncover potential factors behind the team's inability to secure a trophy.

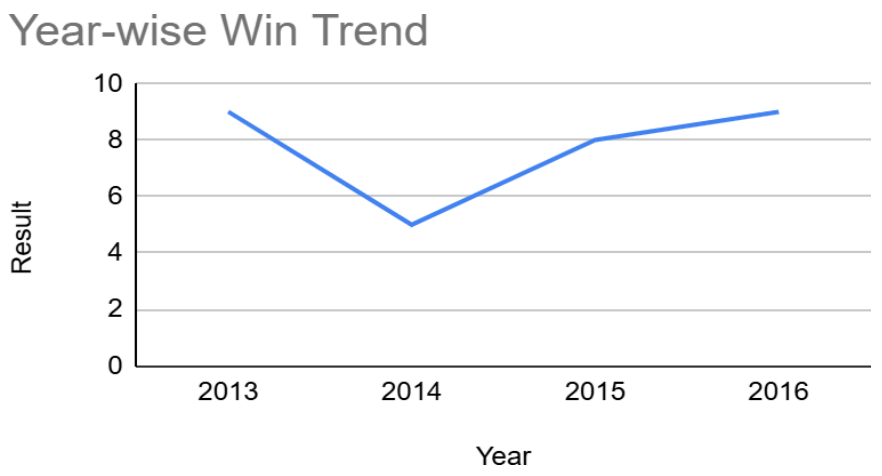
```sql
-- QUERY --
WITH CTE AS (
    SELECT
        m.Match_Id,
        v.Venue_Name,
        CASE
            WHEN t.Team_Id = m.Team_1 THEN t2.Team_Name
            ELSE t1.Team_Name
        END AS Opponent,
        CASE
            WHEN m.Match_Winner = t.Team_Id THEN 1
            ELSE 0
        END AS Result,
        m.Match_Date
    FROM matches m
    JOIN venue v ON m.Venue_Id = v.Venue_Id
    JOIN team t ON t.Team_Name = 'Royal Challengers Bangalore'
    JOIN team t1 ON m.Team_1 = t1.Team_Id
    JOIN team t2 ON m.Team_2 = t2.Team_Id
    WHERE m.Team_1 = t.Team_Id OR m.Team_2 = t.Team_Id
            AND m.Match_Winner IS NOT NULL )
SELECT *
FROM CTE
ORDER BY Match_Date;
```

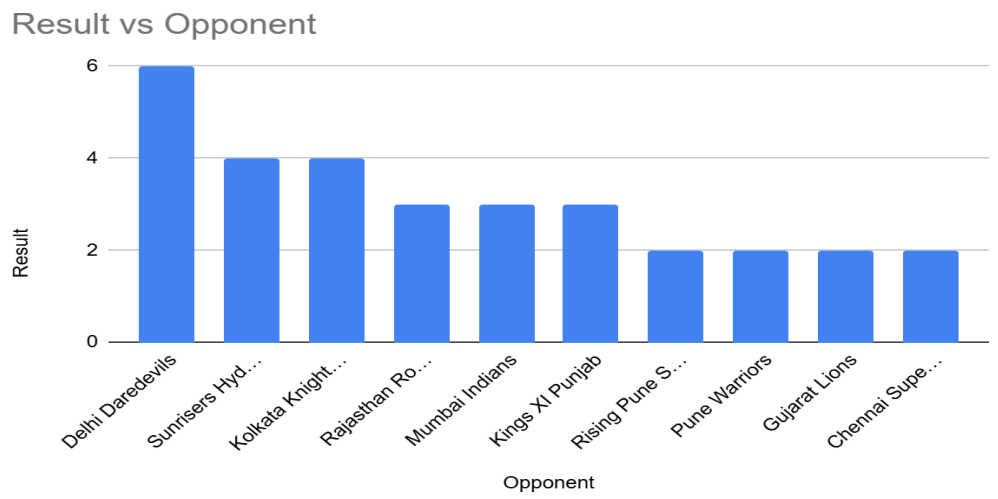
OUTPUT :

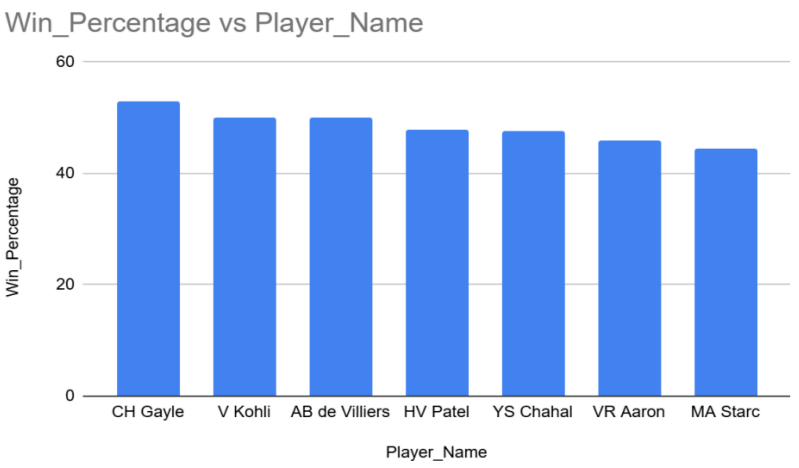
Match_Id	Venue_Name	Opponent	Result	Match_Date
598004	M Chinnaswamy Stadium	Mumbai Indians	1	2013-04-04 00:00:00
598009	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	0	2013-04-07 00:00:00
598053	M Chinnaswamy Stadium	Sunrisers Hyderabad	1	2013-04-09 00:00:00
598013	M Chinnaswamy Stadium	Kolkata Knight Riders	1	2013-04-11 00:00:00
598017	MA Chidambaram Stadium, Chepauk	Chennai Super Kings	0	2013-04-13 00:00:00
598022	M Chinnaswamy Stadium	Delhi Daredevils	1	2013-04-16 00:00:00
598028	M Chinnaswamy Stadium	Rajasthan Royals	1	2013-04-20 00:00:00
598032	M Chinnaswamy Stadium	Pune Warriors	1	2013-04-23 00:00:00
598038	Wankhede Stadium	Mumbai Indians	0	2013-04-27 00:00:00
598041	Sawai Mansingh Stadium	Rajasthan Royals	0	2013-04-29 00:00:00
598047	Subrata Roy Sahara Stadium	Pune Warriors	1	2013-05-02 00:00:00
598069	Punjab Cricket Association Stadium, Mohali	Kings XI Punjab	0	2013-05-06 00:00:00
598059	Feroz Shah Kotla	Delhi Daredevils	1	2013-05-10 00:00:00
598062	JSCA International Stadium Complex	Kolkata Knight Riders	0	2013-05-12 00:00:00

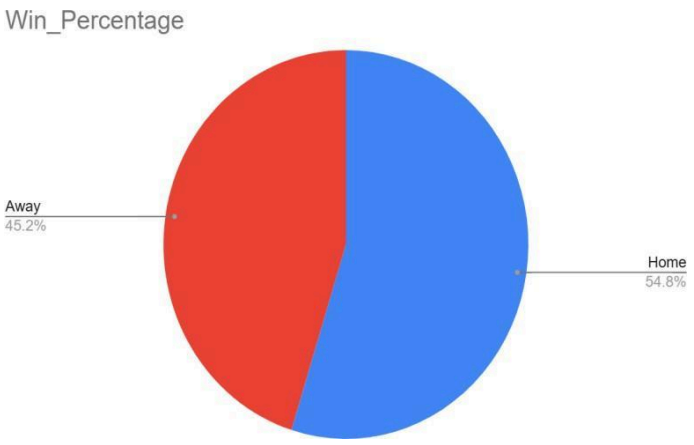
VISUALS :




Year-wise Win Trend
10
8
6
4
2
0
Result
2013
2014
2015
2016
Year


Result vs Opponent
6
4
2
0
Result
Delhi Daredevils
Sunrisers Hyd...
Kolkata Knight...
Rajasthan Ro...
Mumbai Indians
Kings XI Punjab
Rising Pune S...
Pune Warriors
Gujarat Lions
Chennai Supe...
Opponent


Win_Percentage vs Player_Name
60
40
20
0
Win_Percentage
CH Gayle
V Kohli
AB de Villiers
HV Patel
YS Chahal
VR Aaron
MA Starc
Player_Name



INSIGHTS :

- RCB's performance improved over the seasons, with a low in 2014 and noticeable gains in 2015 and 2016, finishing those seasons on a stronger.
- The team's match outcomes are fairly balanced between home and away venues, showing they can perform well in different conditions.
- Key players like CH Gayle, V Kohli, and AB de Villiers have the highest win percentages, underlining their importance in match outcomes.
- Comparing **Home vs Away win percentages** highlights the team's performance consistency across different venues, showing that RCB performs slightly better at home but remains competitive in away matches.

RECOMMENDATIONS :

- Strengthen the middle-order batting to reduce over-reliance on top players like Gayle, Kohli, and de Villiers, ensuring the team can maintain stability even if key batsmen fail.
- Focus on improving away match strategies by analyzing pitch conditions and tailoring game plans, aiming to convert competitive away performances into more consistent wins.
- Develop a balanced bowling attack that combines wicket-taking ability with tight economy across all phases of the game, to complement the batting lineup and maintain pressure on opponents.

10. **How would you approach this problem if the objective and subjective questions weren't given?**

A. If no specific questions were provided, I would approach the analysis in the following way to derive meaningful insights and build effective strategies:

I. **Performance Review:** Analyse last season's results to identify key weaknesses and areas needing reinforcement.

II. **Retain Core Players:** Keep consistent match-winners who perform reliably under high-pressure situations.

III. **Domestic Talent Focus:** Scout in-form domestic players offering both strong performance and cost-effectiveness.

IV. **Middle-Order Stability:** Strengthen the batting core with dependable finishers to prevent collapses.

V. **Death-Over Specialists:** Prioritize bowlers who can execute yorkers and slower balls effectively in the final overs.

VI. **Powerplay Impact:** Target bowlers who consistently take wickets in the first six overs to build early pressure.

VII. **All-Round Balance:** Include multi-skilled players who add flexibility in both batting and bowling departments.

VIII. **Value-Based Selection:** Use data-driven cost-to-performance analysis to ensure every auction buy adds real impact.

11. **In the "Match" table, some entries in the "Opponent_Team" column are incorrectly spelled as "Delhi_Capitals" instead of "Delhi_Daredevils". Write an SQL query to replace all occurrences of "Delhi_Capitals" with "Delhi_Daredevils".**

A. Since the dataset structure includes a *Match* table with an *Opponent_Team* column, it's possible that team names have been entered inconsistently. To ensure data accuracy and uniformity, we need to correct all instances of "Delhi_Capitals" to "Delhi_Daredevils". This update maintains consistency across records and avoids discrepancies during analysis. The SQL query to perform this correction is shown below:

QUERY :

```
-- QUERY --
UPDATE team
SET Team_Name = 'Delhi Daredevils'
WHERE Team_Name = 'Delhi Capitals' ;
```